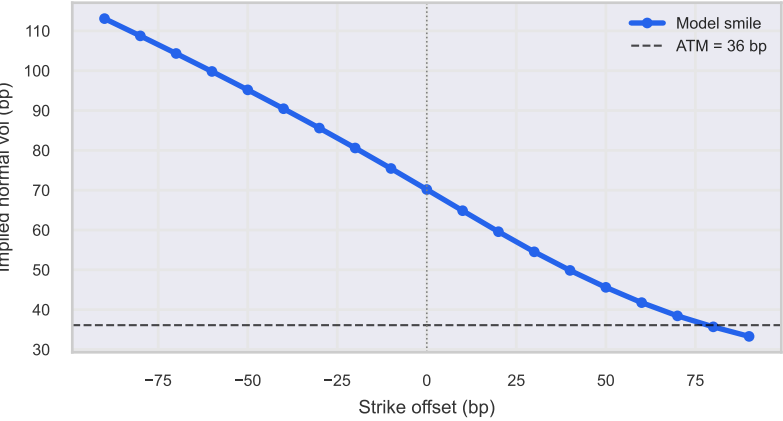
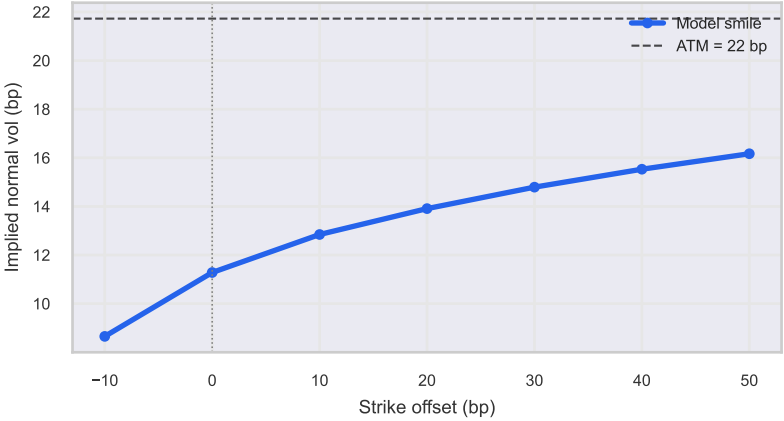


Model-implied Bachelier Vol Smile — Constant MPR  
(rows = rate regime, columns = swaption cell)

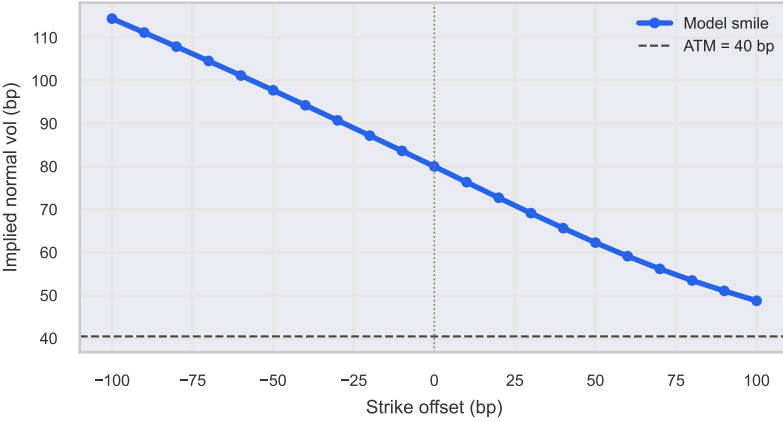
Low rate (2015)  
5Yx1Y ( $F_0 = 1.41\%$ )



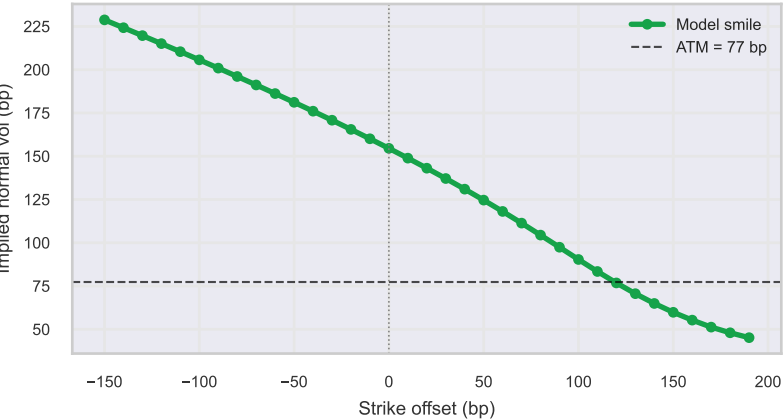
Low rate (2015)  
5Yx5Y ( $F_0 = 1.75\%$ )



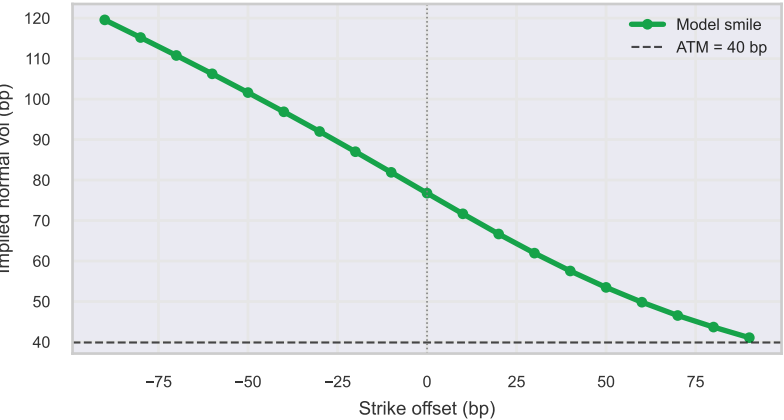
Low rate (2015)  
10Yx10Y ( $F_0 = 2.26\%$ )



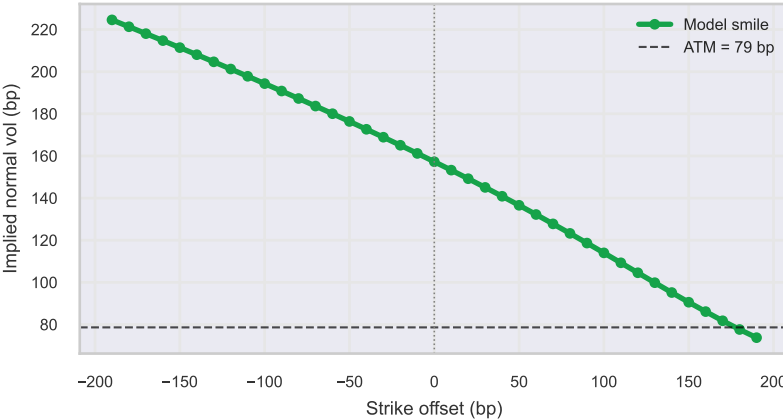
ZIRP (2019)  
5Yx1Y ( $F_0 = 0.50\%$ )



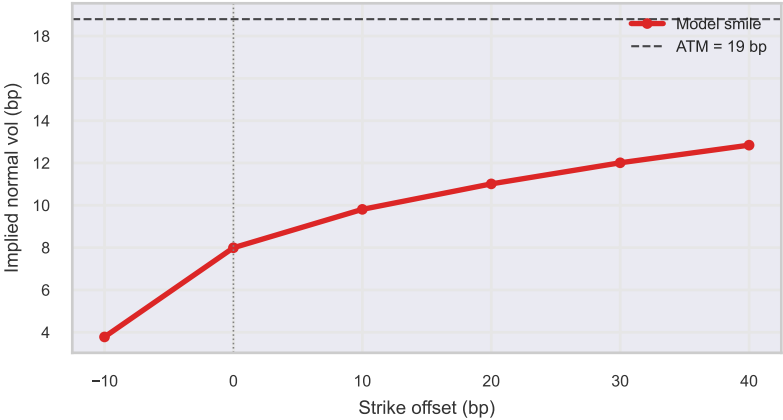
ZIRP (2019)  
5Yx5Y ( $F_0 = 0.69\%$ )



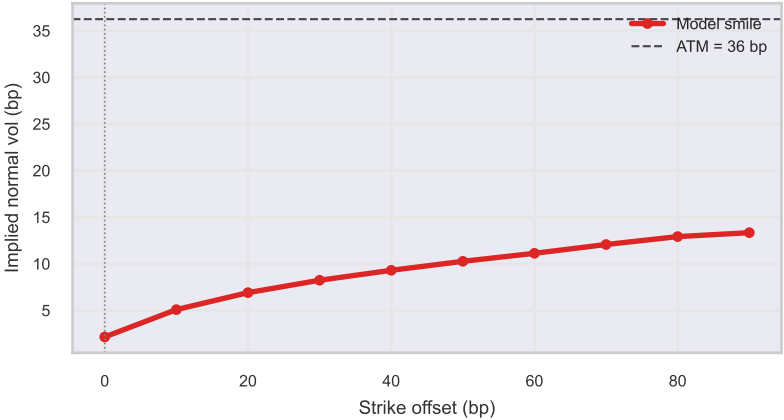
ZIRP (2019)  
10Yx10Y ( $F_0 = 1.03\%$ )



Rate hike (2022)  
5Yx1Y ( $F_0 = 2.45\%$ )



Rate hike (2022)  
5Yx5Y ( $F_0 = 2.46\%$ )



Rate hike (2022)  
10Yx10Y ( $F_0 = 2.36\%$ )

